

MH2100 Calculus III

Midterm I (Solutions)

AY 2023/2024, Semester 1

Quantitative Research Society @NTU

September 2023

Overview

- **Coverage:** parametric curves and tangent lines; limits and continuity; partial and directional derivatives; tangent planes, the Chain Rule, implicit differentiation, and critical points.
- **Historical source:** MH2100 Midterm I, AY2023/2024 Semester 1. Every question below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.

Instructions to candidates (historical, verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(17 Marks)****QUESTION 1.**

- (i) [3 marks] Find parametric equations for the tangent line at the point when $t = 1$ to the curve with parametric equations

$$x = 3t^2, \quad y = 5t, \quad z = 7t^3.$$

- (ii) [3 marks] The cartesian equation $y = ax + b$ describes the line tangent to the curve

$$\begin{cases} x(t) = e^t, \\ y(t) = t^2 + 2t - 7 \end{cases}$$

at the point defined by $t = 0$. Find the values of a and b .

- (iii) [3 marks] Find all points (P, Q) on the following curve where the tangent line is vertical.

$$x = t^3 - 9t + 18, \quad y = \sin(t)$$

- (iv) [3 marks] At the point $(0, \pi, 1)$, find the directional derivative of the function $f(x, y, z) = \sin(xyz)$ in the direction of the vector $\langle 1, 2, 2 \rangle$.

- (v) [5 marks] Find all unit direction vectors $\mathbf{u} = \langle a, b \rangle$ in which the directional derivative of the function $f(x, y) = ye^{-xy}$ at the point $(0, 2)$ has the value 1.

Solution**Method 1: Direct calculation**

- (i) At $t = 1$, the point and velocity are

$$\mathbf{r}(1) = \langle 3, 5, 7 \rangle, \quad \mathbf{r}'(t) = \langle 6t, 5, 21t^2 \rangle, \quad \mathbf{r}'(1) = \langle 6, 5, 21 \rangle.$$

Thus a parametrisation of the tangent line is

$$\boxed{\mathbf{l}(s) = \langle 3, 5, 7 \rangle + s\langle 6, 5, 21 \rangle, \quad s \in \mathbb{R}.}$$

- (ii) The point at $t = 0$ is $(1, -7)$, and

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t+2}{e^t}.$$

Hence the slope is $a = 2$. Substitution of $(1, -7)$ into $y = 2x + b$ gives $b = -9$:

$$\boxed{a = 2, \quad b = -9.}$$

- (iii) A vertical tangent requires $x'(t) = 0$ and $y'(t) \neq 0$. Here

$$x'(t) = 3t^2 - 9, \quad y'(t) = \cos t.$$

Therefore $t = \pm\sqrt{3}$; in each case $\cos t \neq 0$. The corresponding points are

$$\boxed{(P, Q) = (18 - 6\sqrt{3}, \sin \sqrt{3}) \quad \text{or} \quad (18 + 6\sqrt{3}, -\sin \sqrt{3}).}$$

(iv)

$$\nabla f(x, y, z) = \langle yz \cos(xyz), xz \cos(xyz), xy \cos(xyz) \rangle.$$

At $(0, \pi, 1)$ this is $\langle \pi, 0, 0 \rangle$. Since the unit direction is $\langle 1, 2, 2 \rangle / 3$,

$$D_{\mathbf{u}}f(0, \pi, 1) = \langle \pi, 0, 0 \rangle \cdot \frac{\langle 1, 2, 2 \rangle}{3} = \frac{\pi}{3}.$$

(v) Differentiating gives

$$f_x = -y^2 e^{-xy}, \quad f_y = (1 - xy)e^{-xy}, \quad \nabla f(0, 2) = \langle -4, 1 \rangle.$$

The required conditions are $-4a + b = 1$ and $a^2 + b^2 = 1$. With $b = 1 + 4a$,

$$a^2 + (1 + 4a)^2 = 1 \implies a(17a + 8) = 0.$$

Thus

$$\mathbf{u} = \langle 0, 1 \rangle \quad \text{or} \quad \mathbf{u} = \left\langle -\frac{8}{17}, -\frac{15}{17} \right\rangle.$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: point at $t = 1$; 1: tangent vector; 1: correct line.
- (ii) **(3 marks)** 1: point; 1: slope by dy/dx ; 1: both a and b .
- (iii) **(3 marks)** 1: vertical-tangent condition; 1: parameters and nonzero y' check; 1: both points.
- (iv) **(3 marks)** 1: gradient; 1: normalised direction; 1: dot product.
- (v) **(5 marks)** 1: partial derivatives; 1: gradient at the point; 1: directional-derivative equation; 1: unit-vector equation; 1: both vectors.

Question 2**(16 Marks)****QUESTION 2.**

- (i) [4 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y) \rightarrow (1,0)} \frac{\sin(xy)}{x^3y + (x-y)^4}$$

(b)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3y}{x^6 + y^2}$$

- (ii) [5 marks] Find all points (P, Q, R) of the level surface given by

$$x^3 + 3x^2y + 6yz - 3z^2 = 7$$

where the tangent plane has an equation of the form $Ax + Ay = B$ for some constants A and B .

- (iii) [7 marks] Define

$$f(x, y) = \begin{cases} \frac{y^3}{x^2 + y^2}; & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct calculation**

- (i) (a) The denominator tends to $(1 - 0)^4 = 1$, while the numerator tends to $\sin 0 = 0$. Continuity of the quotient near the point gives

$$\boxed{0.}$$

- (b) Along $y = 0$, the quotient is 0. Along $y = x^3$, it is

$$\frac{x^6}{x^6 + x^6} = \frac{1}{2} \quad (x \neq 0).$$

The path limits differ, so

$$\boxed{\text{the limit does not exist.}}$$

- (ii) Let

$$F(x, y, z) = x^3 + 3x^2y + 6yz - 3z^2.$$

The normal to the tangent plane is

$$\nabla F = \langle 3x^2 + 6xy, 3x^2 + 6z, 6y - 6z \rangle.$$

For a plane $Ax + Ay = B$, its normal has equal first two components and zero third component. Thus $z = y$ and

$$3x^2 + 6xy = 3x^2 + 6y \implies y(x - 1) = 0.$$

If $y = z = 0$, the level equation gives $x = \sqrt[3]{7}$. If $x = 1$ and $z = y$, it gives $1 + 3y + 3y^2 = 7$, so $y = 1$ or $y = -2$. Therefore

$$(P, Q, R) = (\sqrt[3]{7}, 0, 0), (1, 1, 1), \text{ or } (1, -2, -2).$$

The gradients at these three points are nonzero. Indeed, their planes are respectively $x + y = \sqrt[3]{7}$, $x + y = 2$, and $x + y = -1$.

(iii) (a) Since

$$|f(x, y)| = \frac{|y|^3}{x^2 + y^2} \leq |y| \rightarrow 0,$$

and $f(0, 0) = 0$. Therefore

$$f \text{ is continuous at } (0, 0).$$

(b) From $f(h, 0) = 0$ and $f(0, h) = h$ for $h \neq 0$,

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 1.$$

(c) If f were differentiable, its derivative would be $L(x, y) = y$. Along $(x, y) = (t, t)$,

$$f(t, t) = \frac{t}{2}, \quad \frac{f(t, t) - L(t, t)}{\sqrt{t^2 + t^2}} = -\frac{t}{2\sqrt{2}|t|},$$

which does not tend to 0. Hence

$$f \text{ is not differentiable at } (0, 0).$$

UNOFFICIAL QRS marking scheme (16 marks)

(i) (4 marks)

(a) (2 marks) 1: nonzero limiting denominator; 1: limit 0.

(b) (2 marks) 1: two valid paths; 1: nonexistence conclusion.

(ii) (5 marks) 1: gradient; 1: plane-normal conditions; 1: branch $y = 0$; 1: branch $x = 1$; 1: all three points.

(iii) (7 marks)

(a) (2 marks) 1: valid bound; 1: continuity conclusion.

(b) (2 marks) 1: f_x ; 1: f_y .

(c) (3 marks) 1: candidate derivative; 1: diagonal-path remainder; 1: non-differentiability conclusion.

Question 3**(17 Marks)****QUESTION 3.**

- (i) [3 marks] Find the second order partial derivative
- w_{xy}
- where

$$w = y \sin(3x^2 - 5y).$$

- (ii) [3 marks] Use the Chain Rule to find
- $\frac{\partial u}{\partial y}$
- at
- $(x, y, z) = (-1, 0, 1)$
- for the function

$$u(p, q, r) = \frac{p^2}{q + r},$$

where

$$p = x + 2y, \quad q = y + 2z, \quad r = x + y + z.$$

- (iii) [3 marks] Assume that
- z
- is uniquely determined in terms of
- x
- and
- y
- by the equation

$$z^3 - xy + yz + y^3 = 2.$$

Find the value of $\frac{\partial z}{\partial y}$ at the point $(1, 1, 1)$.

- (iv) [3 marks] In which directions is the directional derivative of

$$f(x, y) = \sin(xy) - x^2 + xy$$

at $(\pi, 0)$ equal to zero? Write the unit vector for each direction.

- (v) [5 marks] Let

$$f(x, y) = x^3 - y^2 + xy - \frac{x}{2}.$$

- (a) Find all critical points of
- $f(x, y)$
- .

- (b) For each critical point found in Part (a), determine whether or not it is a local max/min or a saddle point.

Solution**Method 1: Direct calculation**

- (i) Differentiating first with respect to
- x
- and then to
- y
- gives

$$w_x = 6xy \cos(3x^2 - 5y),$$

and hence

$$w_{xy} = 6x \cos(3x^2 - 5y) + 30xy \sin(3x^2 - 5y).$$

- (ii) The intermediate values at
- $(-1, 0, 1)$
- are
- $p = -1$
- ,
- $q = 2$
- , and
- $r = 0$
- . Moreover,

$$u_p = \frac{2p}{q + r}, \quad u_q = u_r = -\frac{p^2}{(q + r)^2},$$

while $p_y = 2$, $q_y = 1$, and $r_y = 1$. Therefore

$$\frac{\partial u}{\partial y} = u_p p_y + u_q q_y + u_r r_y = (-1)(2) - \frac{1}{4} - \frac{1}{4} = \boxed{-\frac{5}{2}}.$$

(iii) Differentiate the defining equation with respect to y :

$$3z^2 z_y - x + z + y z_y + 3y^2 = 0.$$

Thus

$$z_y = \frac{x - z - 3y^2}{3z^2 + y},$$

and evaluation at $(1, 1, 1)$ gives

$$\boxed{\frac{\partial z}{\partial y} = -\frac{3}{4}}.$$

(iv)

$$f_x = y \cos(xy) - 2x + y, \quad f_y = x \cos(xy) + x.$$

Thus $\nabla f(\pi, 0) = 2\pi\langle -1, 1 \rangle$. A unit direction $\langle a, b \rangle$ has zero directional derivative precisely when $-a + b = 0$ and $a^2 + b^2 = 1$. Hence

$$\boxed{\mathbf{u} = \frac{1}{\sqrt{2}}\langle 1, 1 \rangle \quad \text{or} \quad \mathbf{u} = \frac{1}{\sqrt{2}}\langle -1, -1 \rangle}.$$

(v) (a)

$$f_x = 3x^2 + y - \frac{1}{2}, \quad f_y = -2y + x.$$

The second equation gives $y = x/2$. Substitution into the first gives

$$6x^2 + x - 1 = 0,$$

so the critical points are

$$\boxed{\left(\frac{1}{3}, \frac{1}{6}\right) \quad \text{and} \quad \left(-\frac{1}{2}, -\frac{1}{4}\right)}.$$

(b) The second derivatives are

$$f_{xx} = 6x, \quad f_{xy} = 1, \quad f_{yy} = -2, \quad D = f_{xx}f_{yy} - f_{xy}^2 = -12x - 1.$$

At $(1/3, 1/6)$, $D = -5 < 0$, so it is a saddle point. At $(-1/2, -1/4)$, $D = 5 > 0$ and $f_{xx} = -3 < 0$, so it is a local maximum. Therefore

$$\boxed{\left(\frac{1}{3}, \frac{1}{6}\right) \text{ is a saddle point, and } \left(-\frac{1}{2}, -\frac{1}{4}\right) \text{ is a local maximum.}}$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: first partial; 1: product and Chain Rules; 1: correct w_{xy} .
- (ii) **(3 marks)** 1: Chain Rule; 1: intermediate values and derivatives; 1: value $-5/2$.
- (iii) **(3 marks)** 1: implicit differentiation; 1: isolates z_y ; 1: evaluates at the point.
- (iv) **(3 marks)** 1: gradient; 1: orthogonality equation; 1: both unit vectors.
- (v) **(5 marks)**
 - (a) **(2 marks)** 1: solves the first-derivative equations; 1: both critical points.
 - (b) **(3 marks)** 1: Hessian discriminant; 1: saddle classification; 1: maximum classification.